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***Differentiating behavioural presentations of biomarker-confirmed Alzheimer's disease from frontotemporal lobar degeneration using the Addenbrooke's Cognitive Examination-III***

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Short Title: Differentiating bvAD from bvFTD using the ACE-III

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**Abstract**
Introduction: Differentiating frontotemporal lobar degeneration (FTLD) from Alzheimer's disease (AD), particularly when presenting with overlapping behavioural symptoms, remains clinically challenging. This study assessed the utility of Addenbrooke's Cognitive Examination-III (ACE-III)-derived ratios in distinguishing these conditions.

Methods: A retrospective cohort of 115 patients (n=52 biomarker-confirmed AD, n=63 FTLD) with overlapping behavioural symptoms was analysed. ACE-III scores and behavioural profiles were examined, and novel ratios were tested.

Results: The novel Phonemic Fluency/Orientation-Memory (PFOM) ratio outperformed existing ACE-III metrics, achieving an area under the curve (AUC) of 0.85 (sensitivity 84.1%, specificity 73.1%) for differentiating FTLD from AD. Notably, longer symptom duration in AD, but not in FTLD, was associated with worsening frontal and cognitive symptoms. Patients with AD were significantly older than those with FTLD.

Discussion: These novel ratios showed robust diagnostic performance, regardless of disease duration, and better reflect FTLD's cognitive profile. They may offer improved clinical utility over traditional ACE-III measures.

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## Introduction

The Addenbrooke's Cognitive Examination (ACE) was developed in 2000 as a bedside cognitive assessment tool to detect early-stage dementia and to differentiate frontotemporal dementia (FTD) from Alzheimer's disease (AD) (1). Since its development it has become widely used to diagnose dementia, with revisions published in 2006 (ACE-R) (2) and 2013 (ACE-III) (3). A technique within the ACE, a ratio derived from dividing the Verbal fluency and Language domain scores by those of attention, Orientation and delayed Memory recall (the VLOM ratio), was proposed as a method to distinguish FTD from AD (1). Since this foundational paper, understanding of the behavioural presentations of AD and Frontotemporal Lobar Degeneration (FTLD) has expanded to include the behavioural/dysexecutive presentation of AD (4), Progressive Supranuclear Palsy frontal presentation (PSP-f) (5), Corticobasal Syndrome frontal-behavioural-spatial subtype (CBS-fbs) (6), and Amyotrophic Lateral Sclerosis-Frontotemporal Dementia (ALS-FTD) (7). Though there is clear overlap in symptomatology, and thus a clinical need to distinguish these presentations of AD and FTLD, no research to date has examined the ACE in differentiating these behavioural subtypes.

Since its original description (1), research on the diagnostic performance of the VLOM ratio in distinguishing FTD from AD has been limited, with only four subsequent studies examining the VLOM in differentiating bvFTD from AD (8, 9, 10, 11). Another study proposed a new ratio, Memory/Verbal Fluency (MVF), with promising preliminary results (12). A further approach using combined logistic regression analyses of ACE and socioeconomic factors has also been proposed, but the complexity of the method would make it challenging to implement in routine clinical settings (10, 13). None of these studies utilised the latest ACE-III, which differs from earlier ACE versions in the language domain, with potential impact for the VLOM ratio. In addition, these past studies have focused on comparisons between bvFTD and typical presentations of AD, raising another potential concern (14). Typical AD presents with memory impairment, whereas bvFTD presents with behaviour and personality changes. Evaluating diagnostic tests in groups with minimal clinical overlap and little diagnostic uncertainty has potentially limited clinical relevance. Thus, the present study explored the clinical utility of the ACE-III, the VLOM, and a range of new simple measures derived from the ACE-III in patients with overlapping behavioural features at initial presentation who were ultimately diagnosed with AD or FTLD (4, 5, 6, 7, 15). The study also examined how disease duration might impact behavioural, cognitive or psychiatric manifestations of these conditions—specifically to test the hypothesis that behavioural symptoms are present from symptom onset in FTLD whereas these symptoms usually emerge later—i.e. after onset of memory symptoms—in AD.

## Methods

*Study design*

A retrospective cohort study was conducted at the Mater Memory and Cognitive Disorders Clinic, a tertiary dementia clinic based in Brisbane, Australia, and The University of Queensland. Patient files were reviewed from a five-year period for a cohort of patients with FTLD and AD who, at initial visit, had informant reported behavioural disturbance that made bvFTD a plausible differential diagnosis (15).

Data were retrospectively collected from patients' electronic medical records, including diagnostic details; follow-up periods; demographic data (sex, age, symptom duration, age at clinic presentation, and age at symptom onset); ACE-III scores; and behavioural and psychological symptom endorsements from an in-house informant behavioural questionnaire (IBQ).

*Inclusion and exclusion criteria*

Patients were included if (i) at the initial clinic presentation they had ≥3 behavioural symptoms within the Rascovsky criteria for bvFTD (criterion A to E) and (ii) on follow up they had either biomarker-confirmed AD or a stable clinical diagnosis within the FTLD spectrum (e.g., bvFTD, PSP-f, CBS-fbs, and ALS-FTD). As detailed in a prior study (16), patients who initially present with bvFTD often on follow-up develop motor symptoms changing their clinical diagnosis to PSP-f, CBS-fbs, or ALS-FTD. This is consistent with the histopathological overlap between these clinical phenotypes (17).

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To ensure diagnostic certainty, all patients were diagnosed according to established research criteria (5, 6, 7, 15, 18, 19). In addition, the bvFTD patients were included if they had histopathological or genetic confirmation, or, met probable bvFTD criteria with functional decline on follow up (mean ± standard deviation, range = 2.5 ± 1.2, 1.0–5.2 years). In the probable bvFTD group, n=5 had negative and n=1 had positive AD biomarkers. The one bvFTD patient with positive AD biomarkers was included as the finding was likely incidental given his age of seventy seven years, noting that about one third of individuals over seventy show amyloid (20). Due to the high prevalence of AD pathology in CBS (21), CBS-fbs patients were only included if they had negative Alzheimer's cerebrospinal fluid (CSF) biomarkers. All AD patients met the research criteria for behavioural/dysexecutive AD and had biomarker confirmation with CSF biomarkers or genetic testing (4). Language presentations of FTLD were not included in the study. The PSP-f and ALS-FTD groups did not require a follow up period, histopathological or genetic confirmation, or exclusion of AD biomarkers due to the high correlation between clinical diagnosis and FTLD histopathology on autopsy (22, 23).

*Patients*

N=115 patients were included in the study: n=63 FTLD and n=52 AD. The FTLD patients included n=25 bvFTD (n=9 definite, n=16 probable), n=4 CBS-fbs, n=13 ALS-FTD, and n=21 PSP-f. Among the AD patients, n=3 had genetic confirmation (all PSEN1 mutations), and the remaining n=49 had CSF biomarker confirmation. To ensure consistency in results between isolated bvFTD cases and the broader FTLD cohort, a subgroup analysis of 'pure bvFTD' was conducted.

To assess the influence of disease stage, patients were stratified into earlier and later disease presentations based on symptom duration of ≤3 or >3 years. Using this definition, n=38 FTLD and n=24 AD patients had earlier, and n=25 FTLD and n=18 AD patients had a later disease stage.

*Defining the behavioural phenotype*

The IBQ was used to document behavioural symptoms at the initial clinic presentation. The IBQ evaluates symptoms across all major syndromes of degenerative dementias, incorporating the frontal behavioural features outlined in the Rascovsky criteria, as well as features of cognitive impairment, functional decline, and psychiatric symptoms. Each item on the questionnaire is rated as present either (0) never, (1) rarely, (2) sometimes, (3) often, or (4) constantly. A score of 2 or more was used as the threshold to define the presence of a behavioural symptom. The IBQ is included in the supplementary material. (See Supplementary Material).

The frontal behavioural symptoms included disinhibition, apathy, lack of empathy, stereotypies and perseveration, and frontal eating behaviours (Q25-36, 38-41). The cognitive items include questions on memory, attention, orientation, visuospatial abilities, language, and executive function (Q1-24). The psychiatric items include questions on psychosis, depression, anxiety, agitation, sleep disturbance, and excessive generosity (Q37, 42-44, 46-53, 55). For each patient, the responses on the IBQ were grouped into these categories to generate a total 'frontal symptom score', 'cognitive symptom score', and 'psychiatric symptom score'.

*Data analysis*

Demographic data, the ACE-III, and, established and novel ratios, were compared between cohorts. The ACE-III total and domain sub-scores were included in the data analysis. The sub-scores included attention and orientation, memory, verbal fluency, language, and visuospatial abilities. Raw scores for delayed name and address recall, phonemic fluency and semantic fluency were also included. The VLOM ratio was calculated by [ verbal fluency + language ] ÷ [ attention & orientation + delayed recall ] (1). The MVF ratio was calculated by [ memory ] ÷ [ verbal fluency ] (12).

*Novel bvFTD ratios*

Three novel cognitive metrics were developed that were hypothesised to possibly have superior performance to the VLOM and MVF: the Phonemic Fluency/Orientation-Memory (PFOM), Phonemic Fluency/Memory (PFM), and



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Phonemic Fluency/Semantic Fluency (PFSF) ratios. The PFOM ratio was calculated as [phonemic fluency raw score] ÷ [attention & orientation sub-score + memory sub-score]. The PFM ratio was calculated as [phonemic fluency raw score] ÷ [memory sub-score]. Finally, the PFSF ratio was calculated as [phonemic fluency raw score] ÷ [semantic fluency raw score].

These ratios were deliberately designed to contrast a domain typically impaired in bvFTD (phonemic fluency) with domains more affected in AD, thereby enhancing separation between the groups. Phonemic fluency raw score, instead of the verbal fluency score, was used due to its specificity to bvFTD and lack of ceiling effect (24). As prior studies found no significant difference in ACE-III language scores between bvFTD and AD (25, 26, 27, 28), the language domain was excluded from these novel ratios. The memory sub-score (scored out of 26) was used instead of the name and address recall item (scored out of 7) to increase the maximum score and reduce the ceiling effect. The PFSF ratio was created due to prior research that showed patients with AD had statistically significant differences between SF and PF scores, whereas patients with bvFTD did not (29). Finally, the ratios were created to remain easy to use, with complex metrics deliberately excluded.

*Statistics*

Statistical analyses were conducted using R statistical software (30). Receiver operating characteristic (ROC) analyses were conducted to assess the discriminative power of the various ACE-III-derived metrics in differentiating FTLD from AD. To assess whether the findings were influenced by the inclusion of a specific FTLD subtype (e.g., PSP-f) a subgroup analysis of pure bvFTD versus AD was conducted.

Sensitivity and specificity were calculated at the optimal cutoff with the highest Youden index. Statistical significance was set at $p<0.05$ with Bonferroni correction for multiple test score comparisons. To ensure interpretive consistency across diagnostic metrics, area under the curve (AUC) values less than 0.5 were mathematically inverted (i.e., recalculated as 1 - AUC), and the corresponding sensitivity and specificity values were interchanged. This adjustment reflects a reversal in classification direction and was performed to ensure that higher scores consistently indicated greater likelihood of FTLD across all metrics. An AUC of 0.5–0.6 was interpreted as a failure, 0.6–0.7 as poor, 0.7–0.8 as fair, 0.8–0.9 as good, and 0.9–1.0 as excellent (31). Pairwise comparisons of AUCs were conducted using the DeLong's test.

Chi-Square ($\chi 2$) was used for nominal data; unpaired t-test for normally distributed numerical data (e.g., frontal symptom score); and Mann-Whitney U test for non-normally distributed numerical data (e.g., symptom duration). Nominal data were presented in proportions (e.g., percentage), normally distributed numerical data were presented as mean ± standard deviation, and non-normally distributed numerical data were reported as median (range). Spearman's rank (one-tailed) was used for non-parametric correlation analyses to assess the relationships between non-normally distributed numerical data. A one-tailed Spearman's rank was used because of an a priori hypothesis that behavioural and cognitive symptoms would increase with disease duration.

*Ethics Statement*

Ethical approval for this retrospective cohort study was granted by the Mater Misericordiae Ltd Human Research Ethics Committee (Approval ID: EC00332), Brisbane, Australia. All analyses were conducted in accordance with the ethical standards of the institutional review board and the principles of the 1964 Declaration of Helsinki and its later amendments.

**Results**

*Demographic details*

Patients with AD were older at symptom onset compared with FTLD patients, and their mean age at clinic presentation was also significantly higher. By comparison, there were no significant differences in symptom duration and sex-distribution between the two groups. The frontal symptom score was significantly higher in the FTLD cohort compared to the AD group. However, there was no significant difference in the cognitive and psychiatric symptom scores between the two groups. (Table 1).



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[Insert Table 1]

*Symptom scores vs disease duration*

In the AD cohort, a longer symptom duration was associated with higher frontal ($\rho$ = 0.24, p = 0.04) and cognitive symptom scores ($\rho$ = 0.30, p = 0.02) but not with the psychiatric symptom score ($\rho$ = 0.07, p = 0.31). In the FTLD cohort, there was no significant correlation between symptom duration and the frontal ($\rho$ = -0.02, p = 0.42), cognitive ($\rho$ = 0.13, p = 0.16), or psychiatric symptom scores ($\rho$ = -0.18, p = 0.08).

*ACE-III group comparisons*

The total ACE-III score was not significantly different between FTLD and AD patients. Among the ACE-III sub-scores, memory, verbal fluency, and phonemic fluency showed the most significant differences, with higher memory scores and lower verbal and phonemic fluency scores observed in FTLD compared to AD. Attention & orientation and delayed memory recall scores were higher in FTLD, but the difference did not remain significant after Bonferroni correction for multiple comparisons. The semantic fluency, language, and visuospatial sub-scores did not differ significantly between the groups. The VLOM, MVF, PFOM, PFM and PFSF ratios were all significantly lower in FTLD compared to AD (Table 2).

[Insert Table 2]

*Differentiating FTLD from AD*

The ACE-III total score, language and visuospatial sub-scores performed close to chance in differentiating the FTLD and AD groups. Phonemic fluency was the best among the sub-scores with a fair test performance, followed by memory and verbal fluency. (Table 3, Figure 1).

The VLOM ratio showed a fair test performance in differentiating FTLD from AD. However, the present cohort showed an optimal cut-off score of <1.76 rather than the originally proposed cut-off score of <2.2 (1), which had a much lower specificity. The novel ratios all outperformed the VLOM, with the PFOM ratio performing the best with a good test performance. The optimal cut-off score was <0.29 with a sensitivity of 84.1% and specificity of 73.1%. Pairwise comparisons of AUCs using DeLong's test showed that PFOM had a significantly better AUC compared to all ACE-III domains and ratios except PFM (z = 1.400, p = 0.162) which had a similar test performance.

[Insert Table 3]

[Insert Figure 1]

*Differentiating the bvFTD subgroup from AD*

In differentiating the pure bvFTD subgroup (n=25) from AD (n=52), the ACE-III total score once again performed similar to chance with an AUC of 0.54. Regarding the sub-scores and ratios, the PFOM ratio had the best test performance with an AUC of 0.76. The optimal cut-off score was <0.28 with a sensitivity of 68.0% and specificity of 73.1%. The PFM and PFSF ratios both had fair test performance (AUC = 0.71 and 0.73). All the other sub-scores, including the VLOM and MVF ratios, performed poorly or similar to chance with an AUC of ≤0.65.

*Differentiating earlier vs later disease presentations*

The PFOM ratio was the most effective measure for differentiating FTLD from AD across both disease stages, demonstrating good test performance in each. The other ratios that incorporated phonemic fluency, the PFM and PFSF ratios, also had consistently fair test performance regardless of disease duration. The VLOM ratio performed significantly worse in later, compared to earlier, disease stages. In comparison, verbal fluency, memory, delayed recall, and the MVF ratio performed better in later, compared to earlier, disease stages. Pairwise comparisons of



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AUCs using DeLong's test for early disease stages showed that PFOM had a significantly better AUC than all ACE-III domains and ratios, except VLOM ($z = 1.15$, $p = 0.25$) and PFM ($z = 1.49$, $p = 0.136$), for which no significant differences were found. For late disease stages the PFOM had a significantly better AUC than all ACE-III domains and ratios, except memory ($z = 1.38$, $p = 0.167$), delayed recall ($z = 1.38$, $p = 0.167$), MVF ($z = -0.509$, $p = 0.610$), PFM ($z = 0.594$, $p = 0.552$) and PFSF ($z = 1.64$, $p = 0.101$). (Table 4).

[Insert Table 4].

**Discussion**

This study demonstrated that ratio-based metrics derived from the ACE-III, that take account of both impaired and relatively preserved cognitive domains, outperformed total or individual subdomain scores in differentiating FTLD from AD. Among these ratios, the novel PFOM and PFM ratios consistently provided the best overall performance. The PFOM had the highest AUC in differentiating FTLD from AD with an AUC of 0.85, sensitivity of 84.1%, and specificity of 73.1%. The PFOM and PFM also outperformed all other metrics when only considering pure bvFTD (i.e. having removed those with PSP-f, CBS-fbs and ALS-FTD); and it was least affected by whether patients presented earlier or later in the course of their disease. Nonetheless this study highlighted the importance of context when interpreting cognitive test results. As AD and FTLD progress, there is a change in clinical phenotype and divergence in cognitive profiles (32). In particular, this study showed that memory and verbal fluency were better at differentiating FTLD from AD in later disease presentations; whereas phonemic fluency and its derived ratios had consistent test performance regardless of symptom duration. The importance of considering disease stage was further emphasized by the observation frontal behavioural scores were significantly correlated with symptom duration in AD but not in FTLD.

The PFOM, which incorporates executive dysfunction via phonemic fluency alongside attention and memory & orientation, appears to better reflect the characteristic cognitive profile of FTLD. This is consistent with previous studies that have identified impairment in phonemic fluency as an early cognitive marker in FTLD (24). In comparison, although the VLOM ratio showed fair performance in differentiating FTLD from AD, it performed poorly for pure bvFTD and was significantly influenced by disease duration. This is consistent with previous studies that have shown the VLOM ratio performed poorly for pure bvFTD (33, 34). This was likely due to the VLOM ratio incorporating the language sub-score—a domain that showed no significant differences between cohorts—and combining both phonemic and semantic fluency, with the latter also failing to show group differences.

The performance of the VLOM has varied between prior studies, depending largely on the comparison cohorts and disease severity. Unlike prior studies that used the VLOM to differentiate bvFTD from a mixed dementia cohort, which showed low sensitivity (33, 34), this study demonstrated a high sensitivity but poor specificity at the same cutoff score. This was likely due to the targeted comparison of FTLD to AD patients with overlapping behavioural features, rather than a broader dementia cohort where bvFTD is not necessarily a diagnostic differential. When comparing the results of this study to prior research of a similar cohort (bvFTD and AD), the results were similar. The ACE-III total score performed similarly in early disease presentations in this study compared to mild disease severity in other studies (AUC = 0.56 vs AUC = 0.5-0.6) (12, 13). In addition, the VLOM ratio had similar test performance in early disease presentations in this study compared to mild disease severity in another study (AUC = 0.79 vs AUC = 0.83) (10). Finally, the MVF ratio also performed similarly in our study compared to prior research (AUC = 0.73 vs AUC = 0.77), with a similar sensitivity and specificity at the same cut-off score (12). These consistencies between the present and past studies offer reassurance that cohorts were comparable, in turn, providing weight to the superiority of the PFOM among the various ratios that were evaluated.

The PFSF was examined because earlier work suggested that semantic and phonemic fluencies were similarly affected in FTD, while semantic fluency was more impaired than phonemic fluency in AD (29). In the present cohort, however, the PFSF showed poor discriminatory ability, and we were unable to replicate the original pattern on which the metric was based. Here, FTD patients performed worse on phonemic than semantic fluency.



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The earlier study included only sixteen FTD patients compared with sixty three in the current series, suggesting that the original finding may have reflected sample variability rather than a consistent group effect.

Despite several strengths, including a relatively large and clinically well-characterised cohort, longitudinal follow-up, and biomarker confirmation of AD pathology, this study has limitations. It was conducted at a single tertiary centre; thus, replication in independent, preferably pathologically confirmed cohorts is needed. Future studies should also examine how these ratios compare with other cognitive tools to determine their broader diagnostic utility.

**Conclusion**

The ACE-III remains a valuable tool for differentiating FTLD from AD. While the ACE-III total score performed consistently similar to chance, specific sub-scores such as phonemic fluency and the novel PFOM ratio significantly enhanced diagnostic accuracy. The newly proposed PFOM and PFM ratios outperformed the VLOM ratio and all other measures across almost all analyses; demonstrating superior sensitivity, specificity, and overall test performance. Unlike the VLOM ratio, which performed poorly in later disease presentations, the PFOM and PFM ratios also reliably distinguished FTLD (and isolated cases of bvFTD) from AD, irrespective of symptom duration. Further validation in independent multi-centre cohorts could establish these ratios as a reliable cognitive marker for accurately differentiating FTLD from AD when there is genuine clinical uncertainty in diagnosis due to an overlap in behavioural symptoms.

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## Statements

*Statement of Ethics*

*This study protocol was reviewed and approved by* Mater Misericordiae Ltd Human Research Ethics Committee (Approval ID: EC00332), Brisbane, Australia. All analyses were conducted in accordance with the ethical standards of the institutional review board and the principles of the 1964 Declaration of Helsinki and its later amendments. As this was a retrospective chart review of de-identified data, the requirement for written informed consent was waived by the approving ethics committee

*Conflict of Interest Statement*

All authors declare no conflicts of interest.

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*Author Contributions*

Conceptualization: J.F., E.A., B.L., T.S., R.A., C.M., P.N.; Methodology: J.F., P.N.; Investigation: J.F., E.A., B.L., T.S., P.N.; Data Curation: J.F., P.N.; Formal Analysis: J.F., T.S., P.N.; Validation: J.F., T.S., P.N.; Visualization: J.F., T.S., P.N.; Writing – Original Draft: J.F., E.A., B.L., T.S., R.A., C.M., P.N.; Writing – Review & Editing: J.F., E.A., B.L., T.S., R.A., C.M., P.N.; Supervision: R.A., P.N..; Project Administration: J.F.; Funding Acquisition: J.F., C.M.; Resources: J.F.; Software: J.F.

## Data Availability Statement

There are ethical restrictions in place for sharing de-identified data. The ethics approval for this project allows for the sharing of de-identified data between institutions included in the study (Mater Misericordiae). The data includes sensitive clinical information relating to the patients' signs, symptoms, medical history, investigations and cognitive assessments. The data reported in the article can be shared with other research groups with appropriate ethical approval by contacting: research.ethics@mater.uq.edu.au

## References

1. Mathuranath PS, Nestor PJ, Berrios GE, Rakowicz W, Hodges JR. A brief cognitive test battery to differentiate Alzheimer's disease and frontotemporal dementia. Neurology. 2000;55(11):1613-20.
2. Mioshi E, Dawson K, Mitchell J, Arnold R, Hodges JR. The Addenbrooke's Cognitive Examination Revised (ACE-R): a brief cognitive test battery for dementia screening. Int J Geriatr Psychiatry. 2006;21(11):1078-85.
3. Hsieh S, Schubert S, Hoon C, Mioshi E, Hodges JR. Validation of the Addenbrooke's Cognitive Examination III in frontotemporal dementia and Alzheimer's disease. Dement Geriatr Cogn Disord. 2013;36(3-4):242-50.
4. Ossenkoppele R, Pijnenburg YA, Perry DC, Cohn-Sheehy BI, Scheltens NM, Vogel JW, et al. The behavioural/dysexecutive variant of Alzheimer's disease: clinical, neuroimaging and pathological features. Brain. 2015;138(Pt 9):2732-49.
5. Höglinger GU, Respondek G, Stamelou M, Kurz C, Josephs KA, Lang AE, et al. Clinical diagnosis of progressive supranuclear palsy: The movement disorder society criteria. Mov Disord. 2017;32(6):853-64.



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6. Armstrong MJ, Litvan I, Lang AE, Bak TH, Bhatia KP, Borroni B, et al. Criteria for the diagnosis of corticobasal degeneration. Neurology. 2013;80(5):496-503.
7. Strong MJ, Abrahams S, Goldstein LH, Woolley S, McLaughlin P, Snowden J, et al. Amyotrophic lateral sclerosis - frontotemporal spectrum disorder (ALS-FTSD): Revised diagnostic criteria. Amyotroph Lateral Scler Frontotemporal Degener. 2017;18(3-4):153-74.
8. Torralva T, Roca M, Gleichgerrcht E, Bonifacio A, Raimondi C, Manes F. Validation of the Spanish Version of the Addenbrooke's Cognitive Examination-Revised (ACE-R). Neurologia. 2011;26(6):351-6.
9. Amaral-Carvalho V, Lima-Silva TB, Mariano LI, de Souza LC, Guimarães HC, Bahia VS, et al. Brazilian Version of Addenbrooke's Cognitive Examination-Revised in the Differential Diagnosis of Alzheimer'S Disease and Behavioral Variant Frontotemporal Dementia. Arch Clin Neuropsychol. 2022;37(2):437-48.
10. Amaral-Carvalho V, Bento Lima-Silva T, Inácio Mariano L, de Souza LC, Cerqueira Guimarães H, Santoro Bahia V, et al. Predicting Dementia Due to Alzheimer's Disease and Behavioral Variant Frontotemporal Dementia Using Algorithms with the Addenbrooke's Cognitive Examination-Revised Subscores Combined with Sociodemographic Factors. Curr Alzheimer Res. 2023;20(5):341-9.
11. Sarasola D, de Luján-Calcagno M, Sabe L, Crivelli L, Torralva T, Roca M, et al. [Validity of the Spanish version of the Addenbrooke's Cognitive Examination for the diagnosis of dementia and to differentiate Alzheimer's disease and frontotemporal dementia]. Rev Neurol. 2005;41(12):717-21.
12. Alexopoulos P, Ebert A, Richter-Schmidinger T, Schöll E, Natale B, Aguilar CA, et al. Validation of the German revised Addenbrooke's cognitive examination for detecting mild cognitive impairment, mild dementia in alzheimer's disease and frontotemporal lobar degeneration. Dement Geriatr Cogn Disord. 2010;29(5):448-56.
13. Amaral-Carvalho V, Bento Lima-Silva T, Inácio Mariano L, Cruz de Souza L, Cerqueira Guimarães H, Santoro Bahia V, et al. Improved Accuracy of the Addenbrooke's Cognitive Examination-Revised in the Diagnosis of Mild Cognitive Impairment, Mild Dementia Due to Alzheimer's Disease and Behavioral Variant Frontotemporal Dementia Using Mokken Scale Analysis. J Alzheimers Dis. 2024;100(s1):S45-s55.
14. Flavell J, Nestor PJ. A systematic review of cognitive and behavioral tools to differentiate behavioral variant frontotemporal dementia from other conditions. PCN Rep. 2024;3(2):e210.
15. Rascovsky K, Hodges JR, Knopman D, Mendez MF, Kramer JH, Neuhaus J, et al. Sensitivity of revised diagnostic criteria for the behavioural variant of frontotemporal dementia. Brain. 2011;134(Pt 9):2456-77.
16. Flavell J, Ahern EGM, Logan B, Shaw TB, Adam RJ, McElligott CAT, et al. Factors associated with true-positive and false-positive diagnoses of behavioural variant frontotemporal dementia in 100 consecutive referrals from specialist physicians. Eur J Neurol. 2025;32(1):e70036.
17. Tartaglia MC, Mackenzie IRA. Recent Advances in Frontotemporal Dementia. Can J Neurol Sci. 2023;50(4):485-94.
18. McKhann GM, Knopman DS, Chertkow H, Hyman BT, Jack CR, Jr., Kawas CH, et al. The diagnosis of dementia due to Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. Alzheimers Dement. 2011;7(3):263-9.
19. Jack CR, Jr., Andrews JS, Beach TG, Buracchio T, Dunn B, Graf A, et al. Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's Association Workgroup. Alzheimers Dement. 2024;20(8):5143-69.
20. Jansen WJ, Janssen O, Tijms BM, Vos SJB, Ossenkoppele R, Visser PJ, et al. Prevalence Estimates of Amyloid Abnormality Across the Alzheimer Disease Clinical Spectrum. JAMA Neurol. 2022;79(3):228-43.
21. Shir D, Pham NTT, Botha H, Koga S, Kouri N, Ali F, et al. Clinicoradiologic and Neuropathologic Evaluation of Corticobasal Syndrome. Neurology. 2023;101(3):e289-e99.
22. Wrigley S, Cullinane PW, Parmera JB, Arca VM, Sifontes Valladares W, Rivera-Sánchez M, et al. Clinical Diagnosis of Progressive Supranuclear Palsy (PSP): A Clinicopathological Comparison of Patients with Confirmed PSP and Clinical Mimics. Mov Disord. 2025;40(10):2116-28.
23. de Carvalho M, Swash M. Diagnosis and differential diagnosis of MND/ALS: IFCN handbook chapter. Clin Neurophysiol Pract. 2024;9:27-38.
24. Hornberger M, Piguet O, Kipps C, Hodges JR. Executive function in progressive and nonprogressive behavioral variant frontotemporal dementia. Neurology. 2008;71(19):1481-8.
25. Bruno D, Slachevsky A, Fiorentino N, Rueda DS, Bruno G, Tagle AR, et al. Argentinian/Chilean validation of the Spanish-language version of Addenbrooke's Cognitive Examination III for diagnosing dementia. Neurologia (Engl Ed). 2020;35(2):82-8.

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26. So M, Foxe D, Kumfor F, Murray C, Hsieh S, Savage G, et al. Addenbrooke's Cognitive Examination III: Psychometric Characteristics and Relations to Functional Ability in Dementia. J Int Neuropsychol Soc. 2018;24(8):854-63.
27. Elamin M, Holloway G, Bak TH, Pal S. The Utility of the Addenbrooke's Cognitive Examination Version Three in Early-Onset Dementia. Dement Geriatr Cogn Disord. 2016;41(1-2):9-15.
28. Parlak MM, Bizpınar Munis Ö, Köse A, Yıldırım C, Ülker CA. The Turkish validity and reliability of Addenbrooke's Cognitive Examination III. Int J Lang Commun Disord. 2025;60(1):e13147-n/a.
29. Rascovsky K, Salmon DP, Hansen LA, Thal LJ, Galasko D. Disparate letter and semantic category fluency deficits in autopsy-confirmed frontotemporal dementia and Alzheimer's disease. Neuropsychology. 2007;21(1):20-30.
30. Team RC. R: A Language and Environment for Statistical Computing. . Vienna, Austria: Foundation for Statistical Computing; 2024.
31. Çorbacıoğlu Ş K, Aksel G. Receiver operating characteristic curve analysis in diagnostic accuracy studies: A guide to interpreting the area under the curve value. Turk J Emerg Med. 2023;23(4):195-8.
32. Johnen A, Bertoux M. Psychological and Cognitive Markers of Behavioral Variant Frontotemporal Dementia-A Clinical Neuropsychologist's View on Diagnostic Criteria and Beyond. Front Neurol. 2019;10:594.
33. Bier JC, Ventura M, Donckels V, Van Eyll E, Claes T, Slama H, et al. Is the Addenbrooke's cognitive examination effective to detect frontotemporal dementia? J Neurol. 2004;251(4):428-31.
34. Yoshida H, Terada S, Honda H, Ata T, Takeda N, Kishimoto Y, et al. Validation of Addenbrooke's cognitive examination for detecting early dementia in a Japanese population. Psychiatry Res. 2011;185(1-2):211-4.

**Figure Legends**

Figure 1. Receiver-operating characteristic (ROC) curves for FTLD vs AD.
ROC curves for (A) ACE-III total and sub-scores, (B) delayed recall, phonemic fluency raw score, and semantic fluency raw score, (C) ACE-III derived ratios.

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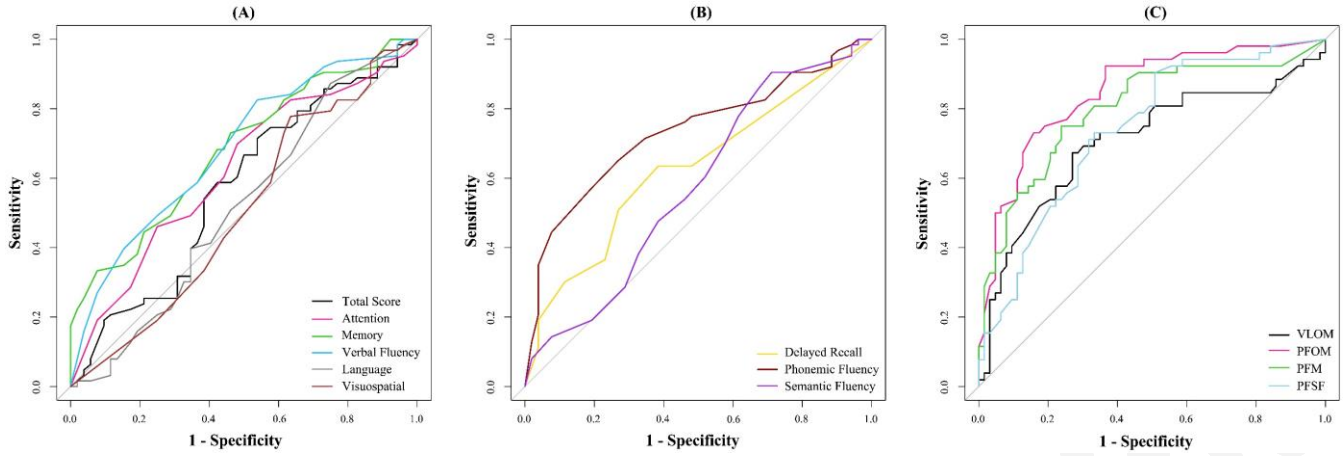


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Table 1. Demographic details and behavioural symptom scores.

| Demographic | AD (n=52) | FTLD (n=63) | Test | P-value |
|---|---|---|---|---|
| Age at symptom onset (years) | 61.8 (42.3–79.5) | 52.2 (28.3–67.8) | U = 493.0 | <0.001 |
| Age at clinic presentation (years) | 65.2 (44.0–82.5) | 56.2 (46.8–69.3) | U = 511.0 | <0.001 |
| Symptom duration (years) | 3.0 (0.5–16.0) | 3.0 (0.2–30.0) | U = 1617.0 | 0.905 |
| Sex (Male:Female) | 1.0:0.73 | 1.0:0.54 | $\chi 2 = 0.658$ | 0.417 |
| Frontal symptom score* | 24.6 ± 9.4 | 30.1 ± 11.7 | t = 2.715 | 0.004 |
| Psychiatric symptom score* | 15.0 (0.0–41.0) | 12.0 (1.0–33.0) | U = 1365.5 | 0.125 |
| Cognitive symptom score* | 48.2 ± 15.0 | 43.3 ± 19.4 | t = -1.486 | 0.650 |

*Derived from the informant behavioural questionnaire (higher scores indicate more severe disturbance).

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Table 2. Comparison of metrics derived from the ACE-III.

| ACE-III | FTLD Median (range) | AD Median (range) | Test (U) | p-value |
|---|---|---|---|---|
| Total Score (/100) | 71.0 (4–94) | 63.5 (4–95) | 1406.5 | 0.193 |
| ***Domain score*** | | | | |
| Attention & orientation ( /18) | 14.0 (1–18) | 12.0 (3–18) | 1237.0 | 0.024 |
| Memory ( /26) | 16.0 (1–26) | 10.0 (0–23) | 1045.5 | ***<0.001*** |
| Verbal fluency ( /14) | 4.0 (0–12) | 6.0 (0–14) | 1031.0 | ***<0.001*** |
| Language ( /26) | 22.0 (2–26) | 23.0 (0–26) | 1577.0 | 0.730 |
| Visuospatial ( /16) | 13.0 (0–16) | 13.0 (0–16) | 1616.5 | 0.903 |
| ***Sub-scores*** | | | | |
| Semantic fluency | 7.0 (0–22) | 8.0 (0–28) | 1388.5 | 0.160 |
| Phonemic fluency | 4.0 (0–16) | 9.0 (0–18) | 877.0 | ***<0.001*** |
| Delayed recall ( /7) | 3.0 (0–7) | 0.0 (0–7) | 1219.5 | 0.014 |
| ***Ratios*** | | | | |
| VLOM | 1.43 (0.57–4.67) | 2.00 (0.25–5.67) | 961.0 | ***<0.001*** |
| MVF | 4.47 (0.00–26.00) | 2.03 (0.00–17.00) | 884.6 | ***<0.001*** |
| PFOM | 0.14 (0.00–0.63) | 0.38 (0.00–1.00) | 493.5 | ***<0.001*** |
| PFM | 0.28 (0.00–1.50) | 0.73 (0.00–7.00) | 661.5 | ***<0.001*** |
| PFSF | 0.56 (0.00–2.33) | 1.00 (0.00–4.00) | 853.3 | ***<0.001*** |

Mann-Whitney U test, with corresponding U-values and p-values provided. ***P-values with bold italics*** survived Bonferroni correction for multiple comparisons.

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Table 3. Test performance of the ACE-III in differentiating FTLD from AD.

| ACE-III | AUC | 95% CI | Cut-off | Youden I | Sensitivity | Specificity |
|---|---|---|---|---|---|---|
| Total score (/100) | 0.571 | 0.464 – 0.678 | ≥58.0 | 0.176 | 71.43 | 46.15 |
| ***Domain score*** | | | | | | |
| Attention & orientation ( /18) | 0.622 | 0.520 – 0.724 | ≥12.5 | 0.218 | 69.84 | 51.92 |
| Memory ( /26) | 0.681 | 0.584 – 0.778 | ≥10.5 | 0.269 | 73.02 | 53.85 |
| Verbal fluency ( /14) | 0.685 | 0.588 – 0.782 | <6.5 | 0.287 | 82.54 | 46.15 |
| Language ( /26) | 0.519 | 0.410 – 0.627 | <25.5 | 0.123 | 87.30 | 25.00 |
| Visuospatial ( /16) | 0.507 | 0.398 – 0.615 | ≥10.5 | 0.143 | 77.78 | 36.54 |
| ***Sub-scores*** | | | | | | |
| Semantic fluency | 0.576 | 0.469 – 0.683 | <12.5 | 0.193 | 90.48 | 28.85 |
| Phonemic fluency | 0.732 | 0.640 – 0.824 | <5.5 | 0.382 | 65.08 | 73.08 |
| Delayed recall ( /7) | 0.628 | 0.526 – 0.730 | ≥1.5 | 0.250 | 63.49 | 61.53 |
| ***Ratios*** | | | | | | |
| VLOM | 0.707 | 0.606 – 0.808 | <1.760* | 0.403 | 73.02 | 67.31 |
| | | | <2.20** | 0.267 | 92.06 | 34.62 |
| MVF | 0.730 | 0.634 – 0.826 | ≥2.93* | 0.456 | 57.14 | 88.46 |
| | | | ≥2.30** | 0.378 | 66.70 | 71.20 |
| PFOM | 0.850 | 0.778 – 0.921 | <0.290 | 0.572 | 84.13 | 73.08 |
| PFM | 0.798 | 0.712 – 0.883 | <0.514 | 0.512 | 76.20 | 75.00 |
| PFSF | 0.739 | 0.649 – 0.830 | <0.686 | 0.397 | 66.70 | 73.10 |

ACE-III: Addenbrooke's Cognitive Examination-III; AUC: Area under the curve; CI: Confidence interval; MVF; PFM; PFOM; PFSF; Youden I: Youden index; VLOM. *optimised cut-off from current series, **cut-off proposed in original publications.

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Table 4. Test performance of the ACE-III-derived metrics when FTLD and AD were stratified into earlier (symptom duration ≤3 years) or later (symptom duration >3 years) disease stages.

| ACE-III | AUC | | Cut-off | | Youden Index | | Sensitivity | | Specificity | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Earlier** | **Later** | **Earlier** | **Later** | **Earlier** | **Later** | **Earlier** | **Later** | **Earlier** | **Later** |
| Total score (/100) | 0.556 | 0.599 | ≥56.5 | ≥47.0 | 0.207 | 0.269 | 73.7 | 88.0 | 47.1 | 38.9 |
| ***Domain score*** | | | | | | | | | | |
| Attention & orientation ( /18) | 0.639 | 0.596 | ≥15.5 | ≥12.5 | 0.235 | 0.236 | 50.0 | 68.0 | 73.5 | 55.6 |
| Memory ( /26) | 0.662 | 0.702 | ≥13.5 | ≥16.5 | 0.302 | 0.313 | 68.4 | 48.0 | 61.8 | 83.3 |
| Verbal fluency ( /14) | 0.672 | 0.710 | <6.5 | <5.5 | 0.257 | 0.371 | 81.6 | 76.0 | 44.1 | 61.1 |
| Language ( /26) | 0.562 | 0.556 | <20.5 | <11.5 | 0.124 | 0.293 | 44.7 | 96.0 | 67.6 | 33.3 |
| Visuospatial ( /16) | 0.526 | 0.530 | ≥10.5 | ≥12.5 | 0.201 | 0.147 | 78.9 | 48.0 | 41.2 | 66.7 |
| ***Sub-scores*** | | | | | | | | | | |
| Semantic fluency | 0.598 | 0.538 | <11.5 | <6.5 | 0.251 | 0.171 | 86.8 | 56.0 | 38.2 | 61.1 |
| Phonemic fluency | 0.729 | 0.728 | <3.5 | <5.5 | 0.359 | 0.513 | 44.7 | 68.0 | 91.2 | 83.3 |
| Delayed recall ( /7) | 0.596 | 0.708 | ≥2.5 | ≥1.5 | 0.203 | 0.498 | 52.6 | 72.0 | 67.6 | 77.8 |
| ***Ratios*** | | | | | | | | | | |
| VLOM | 0.787 | 0.563 | <1.735 | <2.12 | 0.554 | 0.269 | 78.9 | 88.0 | 76.5 | 38.9 |
| MVF | 0.639 | 0.881 | ≥2.93 | ≥1.65 | 0.435 | 0.627 | 55.3 | 96.0 | 88.2 | 66.7 |
| PFOM | 0.857 | 0.843 | <0.285 | <0.235 | 0.604 | 0.609 | 86.8 | 72.0 | 73.5 | 88.9 |
| PFM | 0.808 | 0.790 | <0.323 | <0.514 | 0.543 | 0.538 | 63.2 | 76.0 | 91.2 | 77.8 |
| PFSF | 0.736 | 0.746 | <0.686 | <0.784 | 0.416 | 0.458 | 71.1 | 68.0 | 70.6 | 77.8 |

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